

ReactGhost: A Study in Flight Protocol Trust Boundaries

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Content

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Abstract

In December 2025, React published CVE-2025-55182, a critical Flight protocol vulnerability permitting server-side prototype traversal in React Server Components runtimes. The patch introduced an own-property guard in `getOutlinedModel()` on the server side. This study revisits that fix.

Through static analysis of npm-published artifacts and reproducible runtime tests against shipped releases through React 19.2.5 and Next.js 16.2.3 LTS (April 8, 2026), we document four adjacent findings, labeled ReactGhost-1 through ReactGhost-3 and `$E-eval`, that share the same primitive but land on the unguarded side of the trust boundary, in client bundles, in the ESM variant, and in the server-reference resolution path.

The findings were disclosed to React, Vercel, CERT/CC, and MITRE under standard coordinated practice. Meta posted a formal vendor position on 2026-04-23 declining all four findings as non-vulnerabilities; that position is summarized in the "Vendor Position" section and preserved in full in the CERT/CC VINCE case record. This document records the technical artifacts, the vendor responses received, and the architectural disagreement they surface regarding the placement of the trust boundary in a protocol that crosses a network.

Background

The Flight protocol and the December 2025 patch

React Server Components ship with a custom wire format, Flight, that serializes a tree of components, props, and references for transport between server and client. Both endpoints parse this format into JavaScript values during normal operation.

In December 2025, the React team published CVE-2025-55182, addressing a server-side prototype traversal in the Flight runtime. The patch added an own-property guard to `getOutlinedModel()`.

The research question for this study was narrow: does the same primitive exist elsewhere in the Flight runtime?

Figure 1. The Flight protocol crosses a network. Both endpoints parse attacker-influenceable wire data; the CVE-2025-55182 patch applied to one endpoint.

Findings

Four locations of the same primitive

Each finding below follows a three-part structure: a short summary, the reproduction path, and a brief trust-boundary reading.

Attack flow

RG-1 flaw progression: ESM `requireModule()`

01 · Payload

Flight module reference

`metadata[2]` is set to constructor rather than a real export name.

02 · Runtime

ESM `requireModule()`

The ESM loader resolves the referenced module and reads the requested export.

03 · Flaw

`moduleExports[metadata[2]]`

Direct property access occurs without `hasOwnProperty.call()`.

04 · Traversal

`{}` → Object → Function

Missing own-property validation allows inherited constructor traversal.

The attacker-controlled export name reaches the ESM module export lookup and walks inherited properties because the ESM path lacks the guard present in sibling bundlers.

Summary

The CVE-2025-55182 patch added an own-property guard to `getOutlinedModel()` in the server-side Flight runtime. The ESM variant was not updated.

`requireModule()` in `react-server-dom-esm` indexes its module manifest with a key derived from the Flight payload without verifying that the key is an own property.

Exploitability is scoped to deployments that integrate `react-server-dom-esm` directly (typically meta-framework authors or custom RSC implementations). The package is documented as not intended for direct application use, and Meta's formal position cites this scope in declining CVE assignment.

Reproduction

Performed against the npm-published `react-server-dom-esm` through 19.2.5 by extracting `requireModule`, confirming the absence of an own-property check, and diffing the function body against sibling bundler variants.

Artifact: `react-server-dom-esm@19.2.5` - `esm/react-server-dom-esm-node-loader.js`.

Trust-boundary reading

The same primitive, on the same side of the network, is present in one variant and absent in four.

Attack flow

RG-2 flaw progression: client `getOutlinedModel()`

01 · Payload

`$1:constructor:constructor`

The response references an object chunk followed by prototype property names.

02 · Delivery

Flight stream reaches client

`createFromFetch()` or `createFromReadableStream()` parses network-provided Flight data.

03 · Flaw

`getOutlinedModel()` walks segments

Each segment is resolved with `id = id[reference[i]]` and no own-property guard.

04 · Impact

Function constructor reached

The resolved model can become Function on the application origin.

A Flight reference delivered to the browser is split into path segments and resolved as chained property access inside the client parser.

Summary

`getOutlinedModel()` exists on both the server and client Flight runtimes. The CVE-2025-55182 patch added an own-property guard to the server-side copy; the client-side copy was not changed.

The client copy parses Flight payloads received from the network, allowing the same prototype-walk shape when an inherited property name is referenced.

The realistic delivery vector is not network-position attackers (who already have simpler means to execute script on the victim) but rather server-side reflection or CSRF chains where the legitimate server is coerced into producing a Flight response with attacker-influenced content. CVE-2026-27978 (Next.js pre-16.1.7 Origin:null CSRF) is one documented delivery vector. The Next.js 16.1.7 patch closes this specific vector; the underlying client-side primitive remains, and additional delivery vectors may exist or be discovered.

Reproduction

Static extraction from React Flight client production bundles confirms the absence of an own-property check across variants.

Artifact: `react-server-dom-webpack@19.2.5` - `cjs/react-server-dom-webpack-client.browser.production.js`.

Trust-boundary reading

Both endpoints parse Flight wire data. A parser-hardening change was applied on the server side and not on the client side.

Attack flow

RG-3 flaw progression: server-reference manifest

01 · Request

`$ACTION_ID_constructor`

A multipart form field supplies a prototype key as the action id.

02 · Decode

`decodeAction()` extracts id

The prefix is stripped and the attacker-controlled id enters `loadServerReference()`.

03 · Flaw

`bundlerConfig[id]`

`resolveServerReference()` indexes the manifest without `hasOwnProperty.call()`.

04 · Impact

Invalid module data / DoS

Plain-object manifests, such as Parcel RSC, can resolve prototype data and crash `preloadModule()`.

The `$F` / `$ACTION_ID` path bypasses `getOutlinedModel()` entirely and depends on whether the framework manifest prevents prototype traversal.

Summary

`resolveServerReference()` turns a Flight `$F` server-reference id into a callable through a direct `bundlerConfig[id]` lookup with no own-property guard.

Containment depends on whether the host framework supplies a prototype-less manifest.

Reproduction

Confirmed against the unmodified output of Parcel's official scaffolding command using `react-server-dom-parcel@19.2.5` and `@parcel/rsc@2.16.4`.

```
@parcel/reporter-dev-server: metadata[2].map is not a function
TypeError: metadata[2].map is not a function
```

Trust-boundary reading

The runtime performs lookups against a manifest whose shape it does not control; the relevant invariant is currently framework-level.

Vendor characterization dispute

Meta's 2026-04-23 position asserts that the reported behavior is equivalent to passing any invalid string identifier and does not constitute prototype traversal. The distinction between invalid string IDs and prototype property names is observable in the crash path:

- Invalid string ID: `bundlerConfig[id]` returns `undefined` → catchable "Invalid server action" error → framework can handle gracefully.
- Prototype property: `bundlerConfig[id]` returns inherited `Function` constructor → flows into `preloadModule()` → unhandled `TypeError` on `metadata[2].map is not a function` → Node process exits.

The disagreement is whether the root cause (unguarded property lookup against attacker-controlled input, the same primitive CVE-2025-55182 addressed server-side) represents a runtime-level concern warranting correction, or a framework-level error-handling responsibility. The reader is presented with the observable behavior, the reproduction path against Parcel's own scaffolder, and Meta's architectural counter-position, and is invited to form their own assessment.

Attack flow

\$E flaw progression: development eval path

01 · Prerequisite

Development bundle exposed

The target serves a React Flight development client to reachable users.

02 · Payload

\$E prefix in Flight chunk

The chunk body after \$E contains attacker-controlled JavaScript text.

03 · Flaw

`parseModelString()` eval branch

The dev parser reaches case E and calls `(0, eval)(response)`.

04 · Impact

Direct script execution

Code executes in the browser without needing prototype traversal.

When a development Flight client is exposed, the \$E sigil routes parser input into an eval-based branch that production builds remove.

Summary

The Flight wire format reserves the \$E sigil for an `eval()`-based execution path intended to support development tooling. In a production build the path is dead-code-eliminated; in a development build it is reachable from any payload the client agrees to parse.

Exploitation of this path requires two preconditions: (1) the target application has deployed a React development build to production (a misconfiguration contrary to React's documentation and build tooling defaults), and (2) the attacker has a delivery vector that injects \$E content into a Flight response served to a victim browser (the same class of vector discussed under RG-2). Neither precondition is a React runtime flaw.

This finding is documented as production hardening guidance rather than a runtime vulnerability. Organizations should enforce production-build deployment in CI/CD pipelines and verify via deployment-time scanning (see the YARA rule `React_Development_Build_In_Production` in the Detection Rules section).

Reproduction

Static review confirmed the `eval()` branch is gated behind a development-build environment check and removed in production bundles.

Artifact: `react-server-dom-webpack@19.2.5 - cjs/react-server-dom-webpack-client.browser.development.js`.

Trust-boundary reading

A build artifact does not currently carry an enforceable guarantee about which parser branches it exposes.

Vendor Position

Meta (React core) statement of record

On 2026-04-23, approximately 24 hours before scheduled public disclosure, Meta Security posted a formal position on CERT/CC VU#692236 declining all four ReactGhost findings as non-vulnerabilities. Meta's full statement is preserved in the VINCE case record.

Summarized per finding:

On ReactGhost-1 (ESM `requireModule`): Meta's position is that `react-server-dom-esm` "contains no published code on npm, and is not meant to be directly used as per the warning in its package.json." Meta confirmed the technical behavior through two separate security engineers (Spencer and Duke) during bug bounty review but declined CVE assignment on scope grounds. The package warning cited by Meta states: "This is intended

to be integrated into meta-frameworks. It is not intended to be imported directly." The README adds: "Use it at your own risk."

On ReactGhost-2 (client getOutlinedModel): Meta's position is that exploitation "requires an attacker to have complete control over the flight bytes sent by the server. Attackers in that position can already cause more harm than the impact you describe in your report."

On ReactGhost-3 (resolveServerReference): Meta disputes the prototype-traversal characterization. Meta's direct quote: "We disagree that this is a vulnerability in React, react-server-dom-parcel-server exports a function loadServerAction which you need to call from your application: `const action = await loadServerAction(id)`. If that ID doesn't exist, we throw an 'Invalid server action'. In your PoC you're passing `constructor#x` which will be rejected as a `TypeError`. This doesn't match your description of 'ReactGhost 3' at all as you're framing this as ProtoType traversal, which isn't happening here at all. You could replace `constructor#x` with `invalid-server-id` and you'll observe the same behavior. The solution here is to handle errors properly, which is the developer's responsibility."

On \$E-eval: Meta's position is that the attack model is circular. Meta's direct quote: "This again, similar to ReactGhost-2, requires a bad actor to be in control of the flight bytes stream. This attack model is circular: if an attacker already controls the data stream (server or MITM), they can inject arbitrary JavaScript through countless mechanisms - the \$E prefix is just one of many, and is only exposed in DEV builds, which should never be exposed in production."

Researcher response to Meta's RG-3 characterization

Meta's assertion that invalid string IDs and prototype properties produce identical behavior is technically incomplete. The distinction is material to the finding:

- An invalid string ID (e.g., `invalid-server-id`) causes `bundlerConfig[id]` to return `undefined`, which surfaces as a catchable "Invalid server action" error that frameworks can handle gracefully.
- A prototype property (e.g., `constructor`) causes `bundlerConfig[id]` to return the inherited `Function` `constructor`, which flows into `preloadModule()` and produces an unhandled `TypeError` at `metadata[2].map is not a function` because the returned `Function` `constructor` has no `.chunks` metadata structure.

The observable effect is an unhandled exception that exits the Node process (confirmed in 31ms against the unmodified output of Parcel's official `npm create parcel react-server` scaffolding command using current published npm releases). The crash path is distinct from normal invalid-ID error handling.

Meta's architectural position (that framework-level error handling is the appropriate mitigation layer) has merit. The disagreement is whether the root cause (unguarded property lookup against attacker-controlled input) represents a runtime-level primitive that warrants correction symmetrical to the CVE-2025-55182 patch, or a developer-responsibility error-handling issue that does not.

CVE assignment status

As of public disclosure on 2026-04-24: no CVE identifiers are assigned for RG-1, RG-2, RG-3, or \$E-eval. The coordination record under VU#692236 is the authoritative source.

Reader's disposition

This disclosure presents the technical findings with full reproduction artifacts. Meta's formal position is summarized above and preserved in full in the VINCE case record. Readers are encouraged to evaluate the technical evidence on its merits and form their own assessment of whether the architectural disagreement between independent researcher and vendor should resolve in favor of runtime-level hardening or framework-level containment.

Trust model

The placement of the trust boundary

The Flight protocol crosses a network. Both endpoints parse attacker-influenceable wire data into JavaScript values.

The narrow technical observation is that an own-property check is inexpensive and applies symmetrically across the locations described.

Affected versions and upstream position

Where the guards are present, and where they are not

All components listed below ship the client-side `getOutlinedModel()` from React's Flight client runtime without the own-property guard. The server-side variant was patched for CVE-2025-55182 (December 2025); the client-side copy was not modified by that release. No subsequent vendor release has addressed the client-side ReactGhost findings as of release.

`resolveServerReference()` carries no `hasOwnProperty` guard in any bundler variant. The `$F` server reference path traverses `loadServerReference()` → `resolveServerReference()` → `preloadModule()` → `requireModule()` without invoking `getOutlinedModel()`, so the CVE-2025-55182 fix does not apply along this code path. Parcel RSC is reachable via this path (server DoS from a single unauthenticated request).

Coordinated via CERT/CC VU#692236 (VRF#26-03-PNMHP). CISA Vulnerability Response and Coordination is assigned as case coordinator. Public disclosure: April 24, 2026. This matrix reflects vendor positions as of release.

Affected components

| Component | Affected Packages | Affected Versions | Server-Side CVE-2025-55182 | Client-Side ReactGhost | Server-Side ReactGhost | Notes | | React Flight Client (webpack) | react-server-dom-webpack | All versions through 19.2.5 (April 8, 2026) | Patched (19.0.1+)

| Defense-in-depth | RG-3 (mitigated by Next.js manifest) | Primary target - ships in Next.js
 | | | React Flight Client (turbopack) | react-server-dom-turbopack | All versions through
 19.2.5 (April 8, 2026) | Patched (19.0.1+) | Defense-in-depth | RG-3 (mitigated by Next.js
 Turbopack mode) | | | React Flight Client (parcel) | react-server-dom-parcel | All versions
 through 19.2.5 (April 8, 2026) | Patched (19.0.1+) | Defense-in-depth | Exploitable - server
 DoS confirmed | Plain {} manifest | | | React Flight Client (ESM) | react-server-dom-esm |
 All versions through 19.2.5 (April 8, 2026) | Defense-in-depth | Defense-in-depth | N/A
 (string config) | ESM bundler - hasOwnProperty missing on requireModule() | | | Next.js App
 Router | next | 13.4+ through 16.2.3 | Patched (see below) | Defense-in-depth | Mitigated |
 Largest consumer (Object.create(null) + Proxy) | | | React Router (RSC mode) | react-router
 | 7.x with RSC APIs | Patched via React upgrade | Defense-in-depth | Mitigated (via Vite
 RSC Proxy) | Uses @vitejs/plugin-rsc | | | Shopify Hydrogen | @shopify/hydrogen | 2024.x+ |
 Patched via React upgrade | Defense-in-depth | Mitigated (via React Router/Vite) | Uses
 React Router 7, not RSC direct | | | Waku | waku | All versions | Patched via React upgrade |
 Defense-in-depth | Mitigated (via Vite RSC Proxy) | Uses @vitejs/plugin-rsc | | | Vite RSC
 Plugin | @vitejs/plugin-rsc | All versions | Patched via React upgrade | Defense-in-depth |
 Mitigated (Proxy manifest) | Track Vite releases | | | Parcel RSC | @parcel/rsc | All versions |
 Patched via React upgrade | Defense-in-depth | Exploitable - server DoS confirmed | Plain
 {} manifest (DoS - RG-3) | | | Redwood SDK | rwsdk | All versions | Patched via React
 upgrade | Defense-in-depth | Mitigated (null manifest) | Track Redwood releases | | | Expo
 RSC | Expo with RSC support | RSC-enabled builds | Patched via React upgrade | Defense-
 in-depth | Unverified | Track expo.dev/changelog | | | Custom RSC implementations | Any
 react-server-dom-* | All versions | Varies | Defense-in-depth | Varies - check manifest | If
 manifest is plain {}: vulnerable | |

April 8, 2026 - React 19.2.5 / 19.1.6 / 19.0.5 (Flight security release)

PR #36236 ("*[Flight] Add more cycle protections*") shipped as a coordinated security release across three maintenance branches. The Dependabot advisory describes the release as fixing *"security vulnerabilities in Server Functions."* A byte-level diff of the production builds shows the release modified `extractIterator`, `extractMap`, and `extractSet` in `ReactFlightClient.js`, removing intermediate `response` variable assignments so that the iterator or collection is not tracked by the cycle-detection logic. The release does not modify the code paths covered by the ReactGhost findings. `resolveServerReference()` is byte-identical to 19.2.4 across all bundler variants and all build targets (server.node, server.edge, server.browser, client.node, client.edge, client.browser). The `hasOwnProperty` guard described in this document is absent in the shipped variants on this release line.

Complete CVE and finding reference

| Identifier | Severity | CVSS | Description | Status | | | CVE-2026-27978 | MEDIUM | 4.3 |
 Next.js Server Actions: Origin:null CSRF bypass via sandboxed iframe opaque contexts |
 Patched (Next.js 16.1.7, 2026-03-17) - Assigned by Vercel Open Source as CNA. Patches
 one delivery vector for ReactGhost-2; the underlying client-side primitive remains
 unpatched. | | | CVE-2026-23869 | - | - | Next.js April 8 security release covering

backported bug and security fixes | Patched (Next.js 16.2.3, 2026-04-08) - Does not address ReactGhost client-side primitives; getOutlinedModel in bundled React Flight client is unchanged. | | CVE-2025-55182 | CRITICAL | 10.0 | Server-side RCE via Flight protocol prototype traversal | Patched (React 19.0.1+) | | CVE-2025-66478 | CRITICAL | 10.0 | Next.js-specific variant (merged into CVE-2025-55182) | Patched (Next.js per above) | | CVE-2025-55184 | HIGH | 7.5 | DoS via infinite loop in Flight deserialization | Patched (React 19.0.4+) | | CVE-2025-67779 | HIGH | 7.5 | DoS additional variant | Patched (React 19.0.4+) | | CVE-2026-23864 | HIGH | 7.5 | DoS additional variant (January 2026) | Patched (React 19.0.4+) | | CVE-2025-55183 | MEDIUM | 5.3 | Server Function source code exposure | Patched (React 19.0.3+) | | ReactGhost-1 | HIGH (scope-qualified) | 9.8 (in-scope deployments) | ESM requireModule() missing hasOwnProperty guard (server-side RCE, within react-server-dom-esm deployments only) | UNPATCHED through 19.2.5. Meta confirmed technical behavior through two security engineers, declined CVE citing package-scope warning (not intended for direct use). Meta formal position 2026-04-23: non-vulnerability on scope grounds. | | ReactGhost-2 | MEDIUM (defense-in-depth) / HIGH (when chained) | 8.7 (chained with delivery vector) | Client-side getOutlinedModel() prototype traversal. Defense-in-depth standalone. Exploitable when chained with a delivery vector such as CVE-2026-27978 (Origin:null Server Action CSRF). | UNPATCHED through 19.2.5. Meta formal position 2026-04-23: non-vulnerability; "requires attacker to have complete control over the flight bytes sent by the server." | | ReactGhost-3 | HIGH | 7.5 (DoS standalone) | resolveServerReference() unguarded manifest lookup. Parcel RSC confirmed exploitable for unauthenticated single-POST DoS against the vendor's own scaffolding template (31ms wall time). \$F chain bypasses the CVE-2025-55182 getOutlinedModel() fix entirely. | UNPATCHED through 19.2.5. Verified byte-identical to 19.2.4 in the April 8, 2026 release. Meta formal position 2026-04-23: non-vulnerability; disputes prototype-traversal characterization and attributes crash to developer error-handling responsibility. | | \$E-eval | LOW (standalone hardening) / MEDIUM (chained with reflection + misconfiguration) | Not scored standalone | Production hardening recommendation: React development builds must not be deployed to production. The \$E Flight prefix invokes eval() in development builds; chained with a server-side reflection or CSRF vector against a misconfigured deployment, this enables arbitrary script execution. This is a build-configuration finding, not a runtime vulnerability. | UNPATCHED in development builds (by design). Meta formal position 2026-04-23: non-vulnerability; attack model requires attacker control of Flight stream or production deployment of dev build. | |

Scope of the existing patches

Organizations that adopted the versions listed above received fixes for server-side RCE (CVE-2025-55182), denial of service (CVE-2025-55184, CVE-2025-67779, CVE-2026-23864), and source code exposure (CVE-2025-55183).

The following items are not covered by those releases:

- **Client-side prototype traversal (RG-2).** The `getOutlinedModel()` function at `ReactFlightClient.js:2061` is unchanged across the releases above. The same primitive - prototype chain traversal via `{ }["constructor"]["constructor"] → Function` - is present

in the React Flight client bundle shipped by every RSC framework examined. The client-side `fulfillReference()` carries a `hasOwnProperty` guard; `getOutlinedModel()` does not.

- **ESM server-side RCE (RG-1).** The ESM bundler variant of `requireModule()` in `react-server-dom-esm` carries no `hasOwnProperty` guard. The four other bundler variants (webpack, turbopack, parcel, unbundled) carry it. Verified through React 19.2.5.
- **Server reference manifest bypass (RG-3).** `resolveServerReference()` carries no `hasOwnProperty` guard in any bundler variant. The `$F / $ACTION_ID_` execution path does not invoke `getOutlinedModel()`. Parcel RSC generates a plain-object manifest (`{}`) at the action-manifest registration site, and an unauthenticated server DoS is reachable via `$ACTION_ID_constructor#x`. Next.js, Vite RSC, Waku, and React Router contain the primitive at the framework layer using `Object.create(null)` or Proxy manifests.
- **Client-side DoS protections.** The server-side DoS patches added `arraySizeLimit`, `bumpArrayCount`, and cyclic thenable protection. The corresponding limits are not present on the client side. The client Flight parser does not bound array size or nesting depth.
- **Vendor position (2026-04-23).** Meta formally declined all four ReactGhost findings as non-vulnerabilities during coordinated disclosure under CERT/CC VU#692236. Meta's position is preserved in the VINCE case record and summarized in the "Vendor Position" section of this document. No upstream patch is scheduled as of public disclosure. Organizations must rely on compensating controls, framework-level mitigations, detection rules, and build-configuration discipline (see "Compensating controls" below).

April 8, 2026 security release (19.2.5 / 19.1.6 / 19.0.5). PR #36236 shipped as a coordinated Flight security release that hardened a separate code path (iterator/Map/Set extraction). The ReactGhost-1, -2, -3, and `$E-eval` code paths are unchanged in this release. Verified by byte-level diff of the production builds and by live reproduction of `resolveServerReference()` against the shipped 19.2.5 tarball.

The `getOutlinedModel()` change would mirror the existing server-side patch (approximately six lines, after commit 7dc903c). The `resolveServerReference()` change is a two-line `hasOwnProperty.call()` guard. Neither has shipped at the time of publication.

Compensating controls:

- 1CSP headers: `script-src 'self'; object-src 'none'` on all RSC applications. This is the primary compensating control for RG-2 chained exploitation scenarios regardless of underlying primitive status.
- 2Ensure no React development builds are deployed to production (`$E-eval`)
- 3Next.js applications on 16.1.x and earlier: Upgrade to 16.1.7 or later to receive CVE-2026-27978 (Origin:null Server Action CSRF) patch. Note that this does not address the client-side ReactGhost findings - it closes one delivery vector for ReactGhost-2 but not the underlying primitive. Continue to monitor for Origin: null + multipart/form-data + Next-Action header combinations as defense in depth against future variants.
- 4Parcel RSC: Wrap manifest with `Object.assign(Object.create(null), {...})` in build output. This mitigation addresses RG-3 at the framework layer and is functionally equivalent to the mitigations Next.js, Vite RSC, Waku, and React Router already apply. Independent of

upstream position on vulnerability status, this hardening is recommended for any production Parcel RSC deployment.

- 5Parcel RSC: WAF rule to block `$ACTION_ID_constructor`, `$ACTION_ID_toString`, `$ACTION_ID_proto` in form field names
- 6Deploy the ReactGhost Scanner v3.2.0 to identify affected assets
- 7Vendor-position awareness: Meta's formal position (2026-04-23) declines all four ReactGhost findings as non-vulnerabilities. Organizations should incorporate this disposition into their dependency-management posture and threat models. Continued monitoring of upstream React and Parcel releases is recommended; if architectural framing changes in future vendor communications, the compensating-control posture may be revisited.

Source: react.dev/blog/2025/12/03/critical-security-vulnerability-in-react-server-components (updated January 26, 2026)

Coordination: CERT/CC VU#692236 (VRF#26-03-PNMHP) - Active

CISA Vulnerability Response and Coordination assigned as case coordinator

Public disclosure: April 24, 2026

CVE-2026-27978: github.com/vercel/next.js/security/advisories/GHSA-mq59-m269-xvcx (patched in Next.js 16.1.7, 2026-03-17)

Next.js 16.2.3 release notes: nextjs.org/blog (2026-04-08 backported security and bug fixes, CVE-2026-23869 summary at vercel.com/changelog/summary-of-cve-2026-23869)

Reported by Layau Eulizier Jr - reactghost.com

Tooling

Companion scanner

A research artifact accompanies this publication: a Python scanner that codifies the detection logic discussed above. It runs in passive mode (parsing JavaScript bundles for the patterns this study identifies) and active mode (issuing crafted Flight requests against operator-controlled targets).

The scanner is published under MIT license. Released as-is, intended for defensive inventory work and research reproduction.

File `reactghost-scanner.py` Version 3.3.0 · Python 3.9+ · 3,652 lines Date added April 24, 2026 SHA-256 `cdac5ebf78f918b1ec10f81523df35e06756ac7300119668bc2e78f33fb15503`

Download scanner Copy SHA-256

Usage: `python3 reactghost-scanner.py -u https://target` Verify: `sha256sum reactghost-scanner.py`

Detection rules

Reference logic for SIEM, IDS, and endpoint tooling

The following rule sets are provided as reference detection logic. They are intentionally narrow to the patterns described in this study and are published as-is for defensive use.

SigmaSIEMSuricata / SnortNetworkYARAFile / Endpoint

SIEM-agnostic detection rules. Convert with sigmac for Splunk, Elastic, QRadar, or any supported backend.

Copy All

title: Sigma - **match**: CVE-2025-55182 payload pattern

id: f8a2c3d1-9e4b-4a7f-b5c6-d8e1f2a3b4c5

status: experimental

description: >

Detects CVE-2025-55182 (react2shell) exploitation attempts targeting

React Server Components via crafted multipart POST with prototype

traversal payload.

references:

- <https://github.com/assetnote/react2shell-scanner>
- <https://nvd.nist.gov/vuln/detail/CVE-2025-55182>
- <https://kb.cert.org/vuls/id/692236>

author: NetGuard 24/7 LLC

date: 2026-03-06

modified: 2026-04-22

tags:

- attack.initial_access
- attack.t1190
- cve.2025.55182

logsource:

category: webserver

product: any

```
detection:

  selection_method:

    cs-method: POST

  selection_headers:

    cs-header|contains:

      - 'Next-Action'

    cs-content-type|contains:

      - 'multipart/form-data'

  selection_payload:

    cs-body|contains:

      - '__proto__'

      - 'resolved_model'

  condition: selection_method and selection_headers and selection_payload

level: critical

falsepositives:

  - Authorized penetration testing

  - Security scanner activity (Assetnote, ReactGhost Scanner)

---

---

title: Sigma - match: Flight prototype-traversal references

id: a1b2c3d4-e5f6-7890-abcd-ef1234567890

status: experimental
```

description: >

Detects React Flight wire-format payloads containing constructor chain traversal references targeting the unpatched client-side `getOutlinedModel()` **function**. Broadened in v1.1.0 to cover additional prototype chain variants.

author: NetGuard [24/7](#) LLC

date: [2026-03-06](#)

modified: [2026-04-22](#)

tags:

- `attack.execution`
- `attack.tl059.007`
- `finding.reactghost-2`

logsource:

category: webserver

product: any

detection:

selection_request:

cs-body|contains:

- `'constructor:constructor'`
- `'__proto__:constructor'`
- `'constructor:prototype'`
- `'toString:constructor'`

- 'valueOf:constructor'

selection_response:

sc-body|contains:

- 'constructor:constructor'
- '__proto__:constructor'
- 'constructor:prototype'

selection_content_type:

sc-content-type|contains:

- 'text/x-component'

condition: selection_request or (selection_response and selection_content_type)

level: high

falsepositives:

- Authorized security testing

title: Sigma - **match**: Origin:**null** Server Action invocation

id: b2c3d4e5-f6a7-8901-bcde-f12345678901

status: stable

description: >

Detects cross-origin Server Action invocations via Origin:**null**,

which bypasses Next.js CSRF protection. Commonly delivered via

sandboxed iframes **from** attacker-controlled pages. Severity

upgraded to high following CVE-2026-27978 assignment.

references:

- <https://nvd.nist.gov/vuln/detail/CVE-2026-27978>
- <https://github.com/advisories/GHSA-mq59-m269-xvcx>

author: NetGuard 24/7 LLC

date: 2026-03-06

modified: 2026-04-22

tags:

- attack.initial_access
- attack.t1189
- cve.2026.27978
- cwe.352

logsource:

category: webserver

product: any

detection:

selection_method:

cs-method: POST

selection_action:

cs-header|contains:

- 'Next-Action'

selection_origin:

cs-origin:

- 'null'

selection_content:

cs-content-type|contains:

- 'multipart/form-data'

condition: all of selection_*

level: high

falsepositives:

- Privacy-focused browsers that strip Origin headers
- Sandboxed iframes in legitimate application flows

title: Sigma - **match**: ReactGhost scanner user-agent and behavior

id: c3d4e5f6-a7b8-9012-cdef-123456789012

status: experimental

description: >

Detects known vulnerability scanners probing **for** React Server

Components vulnerabilities. May indicate authorized testing or

external reconnaissance.

author: NetGuard 24/7 LLC

date: 2026-03-06

modified: 2026-04-22

tags:

- attack.reconnaissance
- attack.t1595

logsource:

category: webserver

product: any

detection:

selection_ua_reactghost:

cs-user-agent|contains:

- 'ReactGhostScanner'
- 'ReactGhost'

selection_ua_assetnote:

cs-user-agent|contains:

- 'Assetnote'

selection_behavioral:

cs-method: POST

cs-header|contains:

- 'Next-Action'
- 'rsc-action-id'

cs-content-type|contains:

- 'multipart/form-data'

condition: selection_ua_reactghost or selection_ua_assetnote or selection_behavioral

level: medium

falsepositives:

- Authorized vulnerability scanning
- Internal ASM team scanning

title: Sigma - **match**: bulk client-chunk retrieval with action probing

id: d4e5f6a7-b8c9-0123-def0-234567890123

status: experimental

description: >

Detects a single source IP fetching many Next.js JavaScript chunks in a short window concurrent with or followed by action-header probing. Threshold raised **from 15** to **25** in v1.1.0 to reduce FPs on modern first-load behavior.

author: NetGuard 24/7 LLC

date: 2026-03-06

modified: 2026-04-22

tags:

- attack.reconnaissance
- attack.t1595.002

logsource:

category: webserver

product: any

detection:

chunk_fetches:

cs-method: GET

cs-uri|contains: `'/_next/static/chunks/'`

action_probe:

cs-method: POST

cs-header|contains:

- `'Next-Action'`
- `'rsc-action-id'`

timeframe: 5m

condition: (chunk_fetches | count() by src_ip > 25) and action_probe by src_ip

level: medium

falsepositives:

- CDN pre-warming or cache refresh
- Heavy single-session client activity

title: Sigma - **match**: successful CVE-2025-55182 exploit response

id: e5f6a7b8-c9d0-1234-ef01-345678901234

status: experimental

description: >

Detects server responses indicating successful CVE-2025-55182

exploitation or vulnerability confirmation. A 500 response

with E{"digest" in an RSC response body confirms the server

processed the malicious payload.

author: NetGuard 24/7 LLC

date: 2026-03-06

tags:

- attack.initial_access
- attack.t1190
- cve.2025.55182

logsource:

category: webserver

product: any

detection:

selection_status:

sc-status: 500

selection_content:

sc-content-type|contains:

- 'text/x-component'

selection_body:

sc-body|contains:

- 'E{"digest"'

condition: all of selection_*

level: critical

falsepositives:

- None expected - **this** is a high-confidence indicator of vulnerability

title: Sigma - **match**: rsc-action-id header with prototype property

id: f7e8d9c0-b1a2-3344-5566-778899aabbcc

status: experimental

description: >

Detects ReactGhost-3 exploitation via rsc-action-id header

targeting resolveServerReference() in react-server-dom-*

bundles (webpack, turbopack, parcel) at 19.2.5 and prior.

Unauthenticated single-POST DoS - server crashes on

TypeError in preloadModule() and exits.

references:

- <https://reactghost.com/>
- <https://kb.cert.org/vuls/id/692236>

author: NetGuard 24/7 LLC

date: 2026-04-22

tags:

- attack.initial_access

- attack.impact
- attack.t1499.004
- finding.reactghost-3
- cwe.1321

logsource:

category: webserver

product: any

detection:

selection_method:

cs-method: POST

selection_header:

cs-header|re: 'rsc-action-id:\s*
(constructor|__proto__|toString|hasOwnProperty|valueOf|prototype|isPrototypeOf|propertyIs
Enumerable)(#|\$|\s)'

condition: selection_method and selection_header

level: high

falsepositives:

- Authorized RG-3 regression testing
- NetGuard ReactGhost scanner activity

title: Sigma - match: RSC action POST followed by upstream failure

id: 0a1b2c3d-4e5f-6789-abcd-ef0123456789

status: experimental

description: >

Detects RG-3 denial-of-service pattern - POST with prototype

property value in action header, followed within 60s by

5xx or upstream-unreachable **from** the same destination.

Highest-confidence RG-3 exploitation indicator.

references:

- <https://kb.cert.org/vuls/id/692236>

author: NetGuard 24/7 LLC

date: 2026-04-22

tags:

- attack.impact
- attack.tl499.004
- finding.reactghost-3

logsource:

category: webserver

product: any

detection:

action_post:

cs-method: POST

cs-header|contains:

- 'rsc-action-id'

- 'Next-Action'

upstream_failure:

sc-status:

- 502

- 503

- 504

timeframe: 60s

condition: action_post followed by upstream_failure (correlated on dest)

level: high

falsepositives:

- Coincidental backend failures unrelated to the action request

title: Sigma - **match**: Flight \$E prefix in response body

id: 1b2c3d4e-5f6a-789b-cdef-0123456789ab

status: experimental

description: >

Detects Flight wire-format responses containing \$E prefix,

which invokes eval() in development builds. Presence in

production traffic indicates a development build deployed

to production - a critical misconfiguration.

author: NetGuard [24/7](#) LLC

date: 2026-04-22

tags:

- attack.execution
- attack.t1059.007
- finding.e-eval
- cwe.95

logsource:

category: webserver

product: any

detection:

selection_content:

sc-content-type|contains:

- 'text/x-component'

selection_body:

sc-body|re: '\d+:"\d\$E'

condition: all of selection_*

level: critical

falsepositives:

- Intentional development deployment (should not exist in production)

References

- [CVE-2025-55182 - Server-side Flight prototype traversal \(RCE\).](#)

- [CVE-2026-27978 \(GHSA-mq59-m269-xvcx\)](#) - Next.js Server Action Origin:null CSRF.
- [CVE-2026-23869](#) - Next.js 16.2.3 backported security fixes (April 8, 2026).
- [GHSA-66w5-f972-9mqr](#) - ESM requireModule (RG-1).
- [GHSA-3jqv-w3v4-x4x2](#) - Client getOutlinedModel (RG-2).
- [GHSA-2jxf-ccc6-3w59](#) - resolveServerReference manifest (RG-3).
- [GHSA-8v66-x8j3-mvmw](#) - \$E eval + fulfillReference partial patch.
- [GHSA-9xxr-6f35-gj4h](#) - ESM requireModule RCE (standalone).
- [GHSA-84vm-h4jh-3f2v](#) - \$F chain bypasses getOutlinedModel.
- [GHSA-jhfp-32cf-68r9](#) - Parcel RSC DoS via resolveServerReference.
- [CERT/CC VU#692236](#) - Coordinated disclosure record.
- [React Blog](#) - Critical security vulnerability in React Server Components (December 3, 2025; updated January 26, 2026).
- [Zscaler ThreatLabz](#) - React2Shell write-up on CVE-2025-55182 exploitation patterns.
- [Miggo Security](#) - Independent analysis of Flight prototype traversal.
- [Akamai SIRT](#) - Detection guidance for React2Shell exploitation.
- [Unit 42 \(Palo Alto Networks\)](#) - Field telemetry and exploit pattern review.
- [Raven Security](#) - Reproducible PoC and analysis.

About the author

Layau Eulizier Jr is the founder of NetGuard 24/7™ LLC, an independent security research practice. This study was conducted independently and disclosed under standard coordinated practice.

[NetGuard 24/7™](#)

Disclosure timeline

Disclosure timeline and vendor status

Date	Vendor	Submission	Response
2026-02-26	Vercel Open Source (HackerOne)	RG-2 submitted (#3575021)	Closed - Informative (2026-04-04)
2026-02-27	Meta Bug Bounty	RG-1 submitted (#1709144797124986)	Closed - Informative; behavior confirmed (2026-03-09)
2026-02-27	Meta Bug Bounty	RG-2 submitted (#1714973043208828)	Closed - Informative (2026-03-16)
2026-02-27	Meta Bug Bounty	RG-3 submitted (#1709149007124565)	Advanced to engineering review (2026-03-31); demonstration video delivered (2026-04-16); closed - Informative (2026-04-21)
2026-03-05	Vercel Open Source (HackerOne)	\$E-eval + fulfillReference submitted (#3586257)	Open
2026-03-06	Meta Bug Bounty	\$E-eval + fulfillReference submitted (#1715553939817405)	Closed - Not Applicable (2026-03-06)
2026-03-07	MITRE	CVE request submitted	
2026-03-08	CERT/CC (VINCE)	Coordination submission filed (VRF#26-03-PNMHP)	
2026-03-10		VU#692236 assigned	active
2026-03-09	Meta Bug Bounty	ESM requireModule resubmission (#1718015412904591)	

Closed - Not Applicable; behavior confirmed, declined on scope (2026-03-09) | | | 2026-03-09 | GitHub (facebook/react GHSA) | Seven advisories submitted | In triage | | | 2026-03-17 | Vercel Open Source (CNA) | CVE-2026-27978 published (Next.js 16.1.7 - Origin:null CSRF) | Patched - one RG-2 delivery vector addressed; client-side primitive remains | | | 2026-04-08 | React core | React 19.2.5 / 19.1.6 / 19.0.5 security release (PR #36236) | Verified to leave RG-1, RG-2, RG-3, and \$E-eval code paths unchanged | | | 2026-04-08 | Vercel Open Source | Next.js 16.2.3 LTS security release (CVE-2026-23869) | Client-side getOutlinedModel unchanged across all six compiled Flight client bundles | | | 2026-04-17 | Vercel Open Source (HackerOne) | RG-2 parity report against Next.js 16.2.3 LTS submitted (#3679433) | Open | | | 2026-04-22 | Meta Bug Bounty | RG-3 resubmission against canonical Parcel RSC scaffolder (#1754386072600858) | Open | | | 2026-04-23 | Meta (via CERT/CC VINCE) | Formal vendor position requested by CERT/CC coordinator | All four findings declined as non-vulnerabilities; CVE assignment declined; VINCE case record is authoritative | | | 2026-04-24 | Public disclosure | CERT/CC VU#692236 coordinated publication | Published without CVE assignment. | |

Cite this work

BibTeX

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```
@techreport{eulizier2026reactghost,  
  author      = {Eulizier, Layau},  
  title       = {ReactGhost: A Study in Flight Protocol Trust Boundaries},  
  institution = {NetGuard 24/7 LLC},  
  year        = {2026},  
  month       = {April},  
  type        = {Coordinated Disclosure Research Note},  
  number      = {CERT/CC VU\#692236},  
  url         = {https://reactghost.com/}  
}
```

Plain text

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24/7 LLC. Coordinated Disclosure Research Note. CERT/CC VU#692236.  
https://reactghost.com/
```